

? CryAnalyze: The Complete Project Blueprint

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1. Project Overview & Architecture

CryAnalyze is an AI-powered infant cry triage system. It records audio, processes it via a Machine Learning pipeline, and returns a plain-language diagnosis with medical escalation alerts.

Tech Stack:

- Machine Learning: librosa, TensorFlow / PyTorch
 - Backend: FastAPI (Python)
 - Frontend: React via Vite, TailwindCSS
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2. Phase 1: Building the Backend (From Scratch)

The backend handles the heavy lifting: receiving audio, extracting audio features, and returning the prediction.

Step 1: Environment Setup

1. Create a project folder and navigate into it.
2. Create a Python Virtual Environment: `python -m venv venv`
3. Activate it: `.\venv\Scripts\activate` (Windows)
4. Install dependencies: `pip install fastapi uvicorn python-multipart librosa numpy pydantic tensorflow`

Step 2: The ML Pipeline (`ml\_pipeline.py`)

This script uses librosa to convert the raw audio .wav file into a Mel-spectrogram, which is the standard way to process audio for deep learning.

- Action: Write a function `process_audio(file_path)` that loads the audio and extracts features.
- Integration: Load your trained .h5 CNN model and pass the extracted features into `model.predict()`.

Step 3: The API Server (`main.py`)

- Action: Initialize a FastAPI app and enable CORS (so the React app can communicate with it).
- Action: Create a POST `/analyze` endpoint that accepts an `UploadFile`, saves the audio locally, passes it to `process_audio()`, and returns the JSON result.

Step 4: Run the Backend

Run the server using: `uvicorn main:app --reload --host 0.0.0.0 --port 8000`.

3. Phase 2: Building the Frontend (From Scratch)

The frontend is the "live, explainable AI" dashboard that parents (and judges) will interact with.

Step 1: React Setup

1. Scaffold the app: `npx create-vite@latest frontend --template react`
2. Install UI libraries: `npm install axios lucide-react`
3. Install styling framework: `npm install -D @tailwindcss/postcss tailwindcss`

Step 2: Building the UI (`App.jsx`)

- Capture Mode: Use the browser's MediaRecorder API to capture microphone input, or a standard `<input type="file">` for pre-recorded audio.
- API Integration: Use axios to POST the audio blob to `http://localhost:8000/analyze`.
- Result Display: Build a conditional UI that displays:
 1. The predicted classification (e.g., Hunger, Pain).
 2. A confidence score bar.
 3. An Escalation Alert box (Red for "See a Doctor", Green for "Soothe it").

Step 3: Run the Frontend

Start the React server using: `npm run dev` (Runs on port 5173).

4. Phase 3: Training the AI Model

To move past a prototype, you must train a real model on the Donate-a-Cry corpus.

1. Data Prep: Organize .wav files into folders by class (hungry/, pain/, etc.).
2. Feature Extraction: Write a script looping through all audio files, using `librosa.feature.melspectrogram()` to convert them into 2D numpy arrays (images).
3. CNN Architecture: Build a Convolutional Neural Network (CNN) in TensorFlow/Keras.
 - *Input Layer- -> *Conv2D- -> *MaxPooling2D- -> *Dropout- -> *Dense (Softmax)*
4. Training: Run `model.fit()` on your dataset. Because the dataset is small, use high Dropout (0.5) to prevent overfitting.
5. Export: Save the weights via `model.save('cry_model.h5')`.

5. Phase 4: Pitching & Presentation Guide

To win the Tech4Health track, structure your 3-minute pitch as follows:

- 0:00 - 0:30 (The Hook): Introduce the problem. Talk about exhausted parents at 3 AM and the lack of pediatric access in rural areas. Do not start with technical jargon.
- 0:30 - 1:30 (The Live Demo): Open the React app. Play a crying audio clip into the microphone. Let the judges watch the UI process the audio and display the triage result in real-time.
- 1:30 - 2:30 (The Tech): Explain the architecture. Mention how librosa extracts the acoustic signature and the CNN classifies it. Mention that the app can run fully offline, proving accessibility.

- 2:30 - 3:00 (The USP & Closing): Highlight your Unique Selling Points:
- Zero extra hardware (accessible to low-income families).
- Screening, not diagnosis (clinically safe and responsible).

Q&A Defense Strategy:

If judges ask about medical liability, confidently reply: *"We designed this as a triage aid, not a diagnostic tool. Our UI specifically uses safe escalation language ('Consult a pediatrician') to prevent panic and prioritize infant safety."*